

Deterministic Ground Truth: A 100M Trajectory Physics Corpus for Embodied AI Sovereignty

John Kruze
kruze@aijesusbro.com

March 2026

Abstract

We present G[^]G (Genesis[^]Genesis), a compiled Rust workspace that generates cryptographically sealed physics trajectory training data for embodied AI systems across ten distinct physical domains. Unlike framework-based simulation (MuJoCo, Isaac Sim, Gazebo), every physics substrate in G[^]G is implemented from first principles—force equations derived directly from textbooks, with no abstraction layers. All trajectories are SHA-256 sealed at every timestep, producing a cryptographically verifiable proof chain that allows independent verification of any trajectory from its seed. Running on a single Apple M4 Pro (14-core CPU, 20-core GPU, 24 GB unified memory), G[^]G produced 100,009,600 proof-sealed trajectories and 1,200,449 exported JSON trajectory datasets (28.1 GB) across ten substrates spanning atmospheric entry, underwater navigation, satellite attitude control, soil mechanics, fungal network dynamics, signal propagation, rubble-pile gravity, celestial mechanics, power grid frequency, and deep-time phase dynamics.

We report three empirical discoveries: (1) the Mars EDL suicide burn ignition boundary—a cliff-shaped survival envelope at altitude $\leq v^2/(2(T/m - g)) \times 1.2$ that raises survival rate from 0.7% to 87.27%; (2) a gyroscopic coupling discontinuity in satellite attitude recovery—1-axis models predict full recovery, 3-axis simulation with full Euler coupling recovers only 17.6% of cases at $\omega = 0.5$ rad/s; (3) a percolation phase transition in biological network signal delivery at average edge health ~ 0.4 , matching theoretical lattice percolation thresholds. We additionally document a cross-substrate coupling between fungal network health (Mycelial substrate) and soil bearing capacity (Terran substrate) mediated by glomalin protein, enabling predictive modeling of high-mass robot-soil interaction in bioactive terrain.

Finally, we present an empirical finding on compute architecture: GPU acceleration is mathematically justified for 1 of 10 physics substrates ($O(N^2)$ asteroid rubble-pile gravity). The remaining 9 substrates are single-body sequential physics problems where CPU is the correct architecture. 100M trajectories were generated on 14 CPU cores in 72 hours. A companion deployment on the Internet Computer (ICP) provides publicly queryable physics inference across 9 domains with no server infrastructure and no shutdown vector.

Master proof hashes (100,009,600 trajectories across three sweep generations):

- Sequential (5,509,600): 4b12bf0f...
- Burst (94,500,000):
a9db821f2793cc91bba6a1d07caab
ffce64417edc37759ad6182025b8569fa54
- Personas (999,999): ce481d92...

1 Introduction

1.1 The Training Data Gap

Autonomous systems operating in physical environments face a fundamental gap between available training data and the conditions that actually kill them. Mars landers, underwater vehicles, satellites in eclipse, agricultural robots—all operate at boundary conditions that are systematically underrepresented in existing corpora.

Framework-based simulation (MuJoCo [1], Isaac Sim [2], Gazebo [3]) addresses this partially, but introduces three problems. First, *opacity*: the physics is abstracted behind API calls and configuration files, making root cause analysis of unexpected behavior require understanding framework internals spanning millions of lines of code. Second, *boundary condition blindness*: frameworks optimized for nominal operation produce data clustered around operating centers,

underrepresenting the catastrophic edge cases where autonomous systems fail. Third, *provenance*: there is no cryptographic guarantee that a trajectory was produced by a specific physics model with specific initial conditions.

Recent work on physics-grounded AI training [4, 5, 6] has focused on scaling framework-based simulation across GPU clusters. This paper presents evidence that the scaling assumption is not empirically validated across all relevant physics domains.

1.2 Approach

G[^]G implements ten physics substrates from equations—not frameworks—in compiled Rust, with SHA-256 proof chains sealing every trajectory at every timestep. The corpus is designed around one principle: the data must be traceable. Every force calculation maps to a textbook equation. Every trajectory includes the catastrophic cases. Every timestep is sealed.

The corpus includes 100,009,600 proof-sealed trajectories generated on a single M4 Pro with no cluster and no cloud. This is not presented as a performance benchmark. It is presented as evidence that the physics of embodied AI training data does not require the compute infrastructure currently assumed by the field.

1.3 Contributions

1. A 100M+ trajectory physics corpus across 10 substrates, SHA-256 sealed, publicly queryable on ICP.
2. Empirical discovery of the Mars EDL suicide burn ignition boundary (0.7% to 87.27% survival).
3. Documentation of the gyroscopic coupling discontinuity in satellite attitude recovery.
4. Measurement of percolation threshold in biological network dynamics.
5. Cross-substrate coupling: Mycelial–Terran glomalin dependency for high-mass robot-soil interaction.
6. Empirical evidence against the GPU-first assumption for single-body sequential physics simulation.
7. Demonstration of sovereign physics inference: ICP on-chain deployment with no server and no shutdown vector.

2 System Architecture

2.1 Design Constraints

G[^]G was built under four hard constraints that distinguish it from framework-based simulation:

No external physics libraries. Every equation is hand-rolled from textbook or first-principle derivation. Runtime dependencies: `sha2` (hashing), `serde_json` (serialization), `metal/objc` (Apple GPU), `hex` (encoding). No MuJoCo. No Bullet. No PhysX. No Genesis.

No simulation frameworks. The physics modules *are* the simulation. There is no rendering pipeline, no scene graph, no contact solver abstraction. Each substrate implements its domain equations directly.

Deterministic reproducibility. A Linear Congruential Generator (LCG) with known seed produces identical trajectories on re-execution. Multiplier: 6364136223846793005. Increment: 1442695040888963407. Given the same seed and parameters, any trajectory can be reproduced independently on any machine.

Single machine. All Monte Carlo generation runs on one Apple M4 Pro (14-core CPU, 20-core GPU, 24 GB unified memory). This constraint is not a limitation—it is the empirical basis for the compute architecture finding in Section 6.

2.2 Proof Architecture

Every trajectory is sealed with a cumulative SHA-256 proof chain:

```
ProofChain::new()
    .seed(rng_state)
    .feed_f64(value)    // every timestep,
                       // every physics
                       // variable
    .feed_str(phase)
    .seal()             // final SHA-256
                       state
```

The chain is cumulative: each `feed` operation hashes the new value with the running state. The final `seal` is a function of every value fed—changing any timestep, any initial condition, or any intermediate variable produces a different hash. This creates an auditable chain from initial conditions to final state.

Master proofs chain individual trajectory proofs across corpus sweeps. Three sweep generations produced three master proofs: sequential (5.5M trajecto-

ries), burst (94.5M), and persona-augmented (1M). The composite proof chain covers 100,009,600 trajectories.

3 Physics Substrates

3.1 Mars EDL (Entry, Descent, Landing)

Equations: CO₂ atmospheric density $\rho = 0.020 \cdot e^{-0.00009 \cdot h}$ [kg/m³]; aerodynamic drag $F_d = \frac{1}{2} \rho v^2 C_d A$; Mars gravity $g = 3.721 \text{ m/s}^2$.

Three-phase model: Ballistic entry (Mach 25+) to parachute deployment (subsonic) to retropropulsion. Integration at 1000 Hz Euler.

Survival rate across retropropulsion strategies ($N = 1,377,638$ trajectories):

Strategy	Survival
Hover-and-drop (fixed alt. ignition)	0.7%
Suicide burn (last-moment ignition)	87.27%

The ignition boundary for suicide burn:

$$h \leq \frac{v^2}{2(T/m - g)} \times 1.2 \quad (1)$$

Discovery: The survival envelope is a cliff, not a gradient. Early ignition burns fuel without deceleration benefit; late ignition is unrecoverable. The 1.2 safety factor captures the optimal window. This boundary was not derived analytically—it emerged from 1,377,638 trajectories at 1000 Hz until the energy dynamics revealed the threshold.

3.2 Orbital Attitude Control (The Gyroscopic Discontinuity)

Equations: Euler rotational dynamics $\mathbf{I}\dot{\boldsymbol{\omega}} = \boldsymbol{\tau}_{\text{ext}} - \boldsymbol{\omega} \times (\mathbf{I}\boldsymbol{\omega})$; quaternion propagation $\dot{\mathbf{q}} = \frac{1}{2} \Omega(\boldsymbol{\omega}) \mathbf{q}$; J2–J4 gravitational harmonics; 1PN post-Newtonian corrections.

Recovery rate from initial tumble ($\omega = 0.5 \text{ rad/s}$, $N = 300,041$ trajectories):

Model	Recovery
1-axis (no gyroscopic coupling)	~100%
3-axis (full Euler coupling)	17.6%

The gap is produced by the cross-axis torque term $\boldsymbol{\omega} \times (\mathbf{I}\boldsymbol{\omega})$. In 1D this term is zero. In 3D it creates coupling that drives secondary tumble axes. Rate-gyro-only stabilization (no star tracker) faces a sharp

capture boundary: at $\omega = 0.5 \text{ rad/s}$, 82.4% of trajectories do not recover.

This finding has direct implications for spacecraft autonomy systems that rely on simplified 1-axis models for attitude control design.

3.3 Terran Soil Mechanics

Equations: Boussinesq (1885) subsurface stress:

$$\sigma_z = \frac{3P}{2\pi} \cdot \frac{z^3}{(r^2 + z^2)^{5/2}} \quad (2)$$

Glomalin-modified yield stress: $\sigma_y = (\sigma_{\text{base}} + G_c \cdot k_{\text{glom}}) \cdot f_{\text{moisture}}$.

Mass threshold (Titanhauler, 15 t, dry sand):

Parameter	Value
Contact pressure	94,077 Pa
Dry sand yield stress	29,967 Pa
Pressure/yield ratio	3.14× (soil rejects robot)
Glomalin boost (500 Pa/mg/g)	+41,500 Pa (soil accepts)

Glomalin—a fungal protein produced by arbuscular mycorrhizal fungi—is the coupling mechanism between the Mycelial and Terran substrates. Healthy fungal networks produce glomalin, which raises soil bearing capacity, which enables heavier robots. This cross-substrate dependency is a direct coupling between biological network health and planetary surface robotics.

3.4 Mycelial Network Dynamics (Percolation Threshold)

Equations: Kirchhoff conductance analog: $g = \text{health/length}$; health dynamics: edges above 0.5 strengthen, below 0.5 decay; connected components via union-find.

Measured percolation threshold: average edge health ~ 0.4 .

Above this threshold, signal delivery ratio is high and the network is fully connected. Below it, the network fragments into isolated components and signal delivery collapses to near zero. This is a phase transition, not a gradient. The measured value matches classical 2D lattice percolation theory (~ 0.407 for triangular lattice), providing empirical validation of the Kirchhoff network model.

3.5 Marine Navigation

Equations: Archimedes buoyancy $F_b = \rho_{sw} \cdot g \cdot V$ ($\rho = 1025 \text{ kg/m}^3$); hydrodynamic drag $F_d = \frac{1}{2} \rho |v|^2 C_d A$; depth from pressure $z = (P - P_{\text{atm}})/(\rho g)$.

GPS-denied dead reckoning creates a fundamental navigation asymmetry: vertical position is precise from pressure ($\pm 0.01 \text{ m}$), horizontal position accumulates IMU drift error ($0.02\text{--}0.1 \text{ m/s}$). This asymmetry defines the navigation dynamics problem unique to underwater vehicles. 1000 Hz integration rate makes this the most computationally expensive substrate per trajectory.

3.6 Remaining Substrates

Atheric (signal propagation): Friis path loss + Shannon capacity + SHA-256 frequency hopping. Key finding: clock drift above the synchronization window collapses an N -channel system to $1/N$ effective bandwidth. Channel sovereignty depends on clock sovereignty.

Asteroid rubble pile: $O(N^2)$ Newtonian gravity (32–85 boulders, 1,024–7,225 force pairs). Sole substrate where Apple Metal GPU is authoritative physics engine. MSL compute kernel dispatched at runtime. Softening factor 10^{-4} prevents singularities.

Celestial N-body: Sun–Earth–Mars 4-body gravitational field. Yoshida 4th-order symplectic integrator preserving energy over long timescales. 1PN relativistic corrections.

Plutonian (deep time): 100-year timesteps over 1M years. Models entropy accumulation and coherence collapse—structural reorganization dynamics applicable to software architecture and organizational systems.

Energy (power grid): Swing equation $df/dt = (P_g - P_d) \cdot f_0 / (2HS_{\text{base}})$. Blackout threshold at $f < 59.3 \text{ Hz}$ or $f > 60.7 \text{ Hz}$. IEC wind power curves. 186,060 trajectories/second—fastest substrate.

4 Corpus

4.1 Generation Statistics

Export size: 28.1 GB across 50 files. Hardware: Apple M4 Pro, 14-core CPU, 20-core GPU, 24 GB unified memory. Total sealed compute: ~ 72 hours.

Substrate	Rate	Sealed	Export
Energy	186,060/s	890,000+	26,487
Atheric	15,830/s	55,548,980	599,578
Terran	10,750/s	36,312,728	399,255
Mycelial	760/s	1,932,161	30,671
Orbital	353/s	300,041	24,225
Mars	144/s	1,377,638	25,033
Plutonian	71/s	470,000+	23,928
Celestial	71/s	670,000+	25,140
Marine	11/s	28,452	21,396
Asteroid	9/s	600,000+	24,736
Total		100,009,600+	1,200,449

Table 1: Generation statistics across all 10 substrates.

4.2 Trajectory Format

Each exported trajectory includes: unique ID (substrate prefix + seed hash), outcome score, SHA-256 proof hash, anomaly classification, initial condition snapshot (persona, mass, soil type, failure mode), and full timestep-by-timestep state data. The proof hash can be independently verified by reseeding the LCG with the recorded seed and replaying the physics forward.

4.3 ICP Deployment

Three Rust canisters on ICP mainnet provide public physics inference:

- Backend: `ad7wi-4aaaa-aaaad-aeijq-cai` (9 domains)
- Agent: `qb2yf-mqaaa-aaaak-qwria-cai`
- Frontend: `qax6c-taaaa-aaaad-qjqqy-cai`

The on-chain deployment is not a training corpus. It is a sovereign physics inference engine: compute runs on ICP’s decentralized network with no server, no API key requirement, and no shutdown vector.

5 The Proprioceptive Ghost: Navigation Under Total Sensor Denial

We introduce the term “Proprioceptive Ghost” to describe the navigational state of an autonomous system operating under total external sensor denial (GPS denied, no optical reference, no external timing).

In this state, the vehicle’s only ground truth is its own integration history. Every sensor that connects it to external reference has failed. The vehicle must navigate using accumulated internal state alone: IMU integration for position, rate gyros for attitude, pres-

sure for depth (marine), accumulated timestep for time.

G[^]G’s Marine and Orbital substrates are designed to generate Proprioceptive Ghost scenarios at volume. The Marine substrate generates GPS-denied submarine missions where horizontal position is dead-reckoned entirely from IMU data. The Orbital substrate generates dark-window satellite recovery scenarios where the spacecraft must stabilize using only rate gyros—no star tracker, no external reference.

Key properties of Proprioceptive Ghost trajectories:

1. **Error accumulation is the physics.** IMU drift, gyro bias, pressure noise—these are not artifacts to be filtered. They define the navigation problem.
2. **The recovery boundary is sharp.** The 17.6% orbital recovery rate at $\omega = 0.5 \text{ rad/s}$ is not a smooth function of initial conditions. It is a cliff—small changes in initial tumble axis create dramatic divergence in recovery probability.
3. **Deterministic simulation reveals the boundary.** Only by running many trajectories from diverse initial conditions does the recovery envelope become visible. Single-trajectory simulation produces single data points. Monte Carlo simulation produces the shape of the problem.

The Proprioceptive Ghost is the training scenario for spacecraft autonomy. Vehicles operating beyond the communication horizon—Mars-surface, deep ocean, far from base station—are Proprioceptive Ghosts. Their survival depends on physics models trained on data generated at exactly this boundary.

6 Compute Architecture: The CPU-GPU Balance

6.1 The Industry Assumption

The robotics and embodied AI field operates on an implicit assumption: physics simulation is a GPU problem. NVIDIA Isaac Sim, Genesis, MuJoCo MJX, and DeepMind’s simulation infrastructure are built on data parallelism across GPU threads—thousands of environments running simultaneously. This scales simulator throughput by exploiting parallelism within each timestep.

G[^]G tested this assumption across 10 physics substrates. The result: **GPU is the correct compute architecture for 1 of 10 substrates.**

6.2 GPU Requirements

GPU acceleration requires three conditions to produce net benefit:

1. *Data parallelism:* the same operation applied to many independent data points simultaneously.
2. *Minimal branching:* GPU threads that diverge through if/else or state machines serialize and lose parallelism benefit.
3. *Compute dominance over overhead:* physics compute must exceed GPU dispatch cost (command buffer creation, encoder setup, thread synchronization, memory barriers).

6.3 Substrate Analysis

Substrate	GPU Case	Verdict
Mars	1-body, 3-phase state machine	CPU
Marine	1-body, 1000 Hz, IMU drift	CPU
Orbital	1-body, quaternion, 5+ branches	CPU
Terran	1-body, 4 sequential phases	CPU
Mycelial	Graph, union-find	CPU
Atheric	Discrete event, crypto hopping	CPU
Energy	1-node, swing equation	CPU
Plutonian	State machine, 100-yr steps	CPU
Celestial	N=4, 6 force pairs	CPU
Asteroid	85 bodies, 7,225 pairs	GPU

Table 2: CPU vs. GPU verdict for each substrate.

6.4 The Single-Body Result

Nine of ten substrates are single-body physics problems: one vehicle, one atmosphere; one submarine, one ocean; one satellite, one set of Euler equations. The physics is inherently serial—each timestep depends on the previous. There is no parallelism to exploit within a single trajectory.

The industry scales GPU clusters to run thousands of such simulations in parallel. G[^]G generates them in parallel on CPU (one trajectory per CPU core). For single-body sequential physics, these are equivalent approaches—except CPU has lower overhead per trajectory and higher numerical precision (no f64-to-f32 conversion required for non-GPU substrates).

6.5 The Asteroid Exception

Asteroid rubble pile dynamics meet all three GPU conditions: $O(N^2)$ identical force pairs (data parallel), no branching (every thread computes $F = Gm_1m_2/r^2$), and compute dominates overhead at $N = 85$ (7,225 force pairs per timestep). G[^]G implements this with an Apple Metal compute shader:

- Thread i computes gravitational force from all $j \neq i$ bodies in parallel
- Symplectic kick-drift integration applied per-thread
- Softening factor 10^{-4} prevents singularities at close approach
- Metal unified memory (`StorageModeShared`) eliminates PCIe copy overhead

The CPU physics loop was removed for this substrate. The Metal shader is the physics engine.

6.6 Implication

100M trajectories were generated on 14 CPU cores in 72 hours of compute on a single laptop.

The hyperscaling assumption—that physics simulation for embodied AI training data requires GPU clusters—is not validated by this evidence. For single-body sequential physics, which covers the majority of autonomous system domains (EDL, underwater, satellite, ground robotics), CPU is the correct architecture. GPU clusters add cost and infrastructure without adding physics.

The $16,914\times$ rate spread between Energy (186,060/s) and Marine (11/s) is not a compute bottleneck. It is the physics. A power grid is a single scalar equation. A submarine is 1000 Hz 3D integration with current fields. No GPU scaling changes what the physics costs to compute.

7 Discussion

7.1 What This Corpus Is Not

This corpus is not a claim that all physics simulation should move to CPU. Multi-body contact dynamics, fluid simulation, cloth simulation, and crowd simulation have legitimate GPU use cases. The claim is narrow and empirical: for the specific physics domains relevant to spacecraft autonomy, underwater navigation, and ground robotics, the 10 substrates in G[^]G run correctly and efficiently on CPU.

This corpus is also not a claim that framework-based simulation is wrong. MuJoCo and Isaac Sim produce valid physics. The claim is that framework abstraction makes the physics non-inspectable and the data non-verifiable. For training embodied AI systems that must survive boundary conditions, inspectability and verifiability are not optional.

7.2 What Comes Next

The on-chain ICP deployment demonstrates that physics reasoning can be sovereign—running without permission, without infrastructure dependency, without shutdown vector. The Monte Carlo generator demonstrates that physics data can be cryptographically verified. Together they define a trajectory toward autonomous systems that carry their own physics models, verified from first principles, in sovereign infrastructure.

The thesis: software evolves through physics selection pressure, not human annotation. The corpus is the evidence.

8 Code and Data Availability

Corpus availability: Individual substrate datasets and the full 100M+ trajectory corpus are available for licensing. See website for details.

ICP public inference (free):

- Backend canister:
ad7wi-4aaaa-aaaad-aeijq-cai
- Public frontend: <https://qax6c-taaaa-aaaad-qjyqy-cai.icp0.io>

Website: <https://embodied-reasoning.web.app>

Contact: kruze@aijesusbro.com

Master proof verification: Any trajectory from this corpus can be independently verified by reseeding the LCG with the trajectory’s recorded seed and replaying the physics equations. Three master proofs cover the 100,009,600 trajectory corpus (see Abstract for hashes).

References

- [1] E. Todorov, T. Erez, and Y. Tassa, “MuJoCo: A physics engine for model-based control,” in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2012.

- [2] V. Makoviychuk *et al.*, “Isaac Gym: High performance GPU-based physics simulation for robot learning,” in *NeurIPS Datasets and Benchmarks*, 2021.
- [3] N. Koenig and A. Howard, “Design and use paradigms for Gazebo, an open-source multi-robot simulator,” in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2004.
- [4] N. Rudin *et al.*, “Learning to walk in minutes using massively parallel deep reinforcement learning,” in *Proc. Conf. Robot Learning (CoRL)*, 2022.
- [5] V. Mnih *et al.*, “Human-level control through deep reinforcement learning,” *Nature*, vol. 518, pp. 529–533, 2015.
- [6] OpenAI *et al.*, “Solving Rubik’s Cube with a Robot Hand,” arXiv:1910.07113, 2019.

$G^{\wedge}G$: *Genesis^Genesis — One M4 Pro. No cluster. No permission.*

John Kruze, March 2026

`kruze@aiesusbro.com`